

Theories of Population Growth and Decline



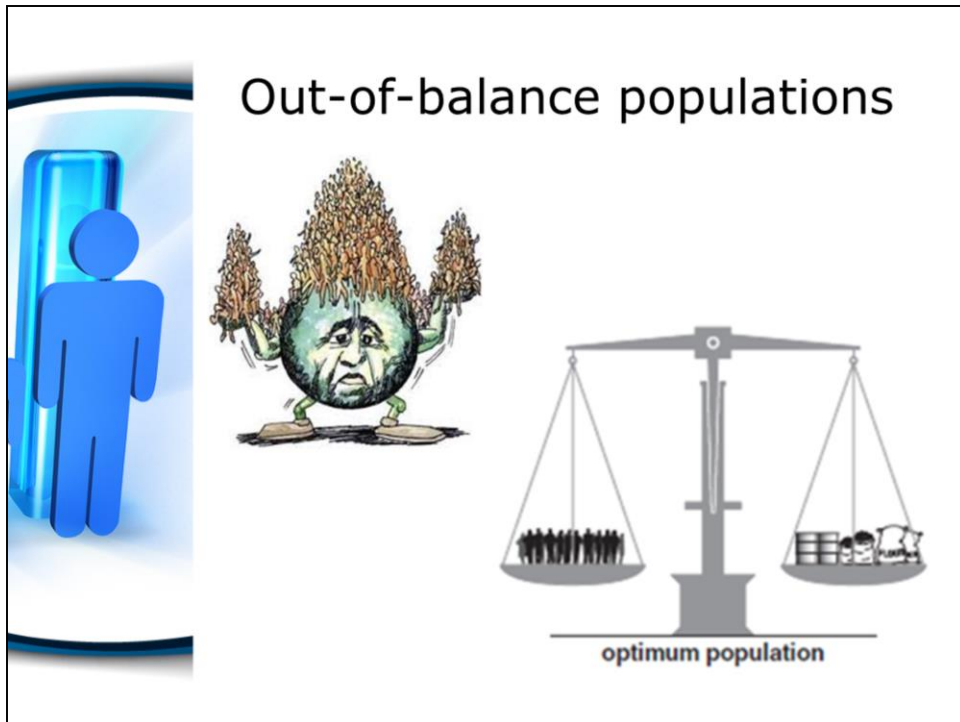
RRHS APHG 2019-2020

Carrying Capacity



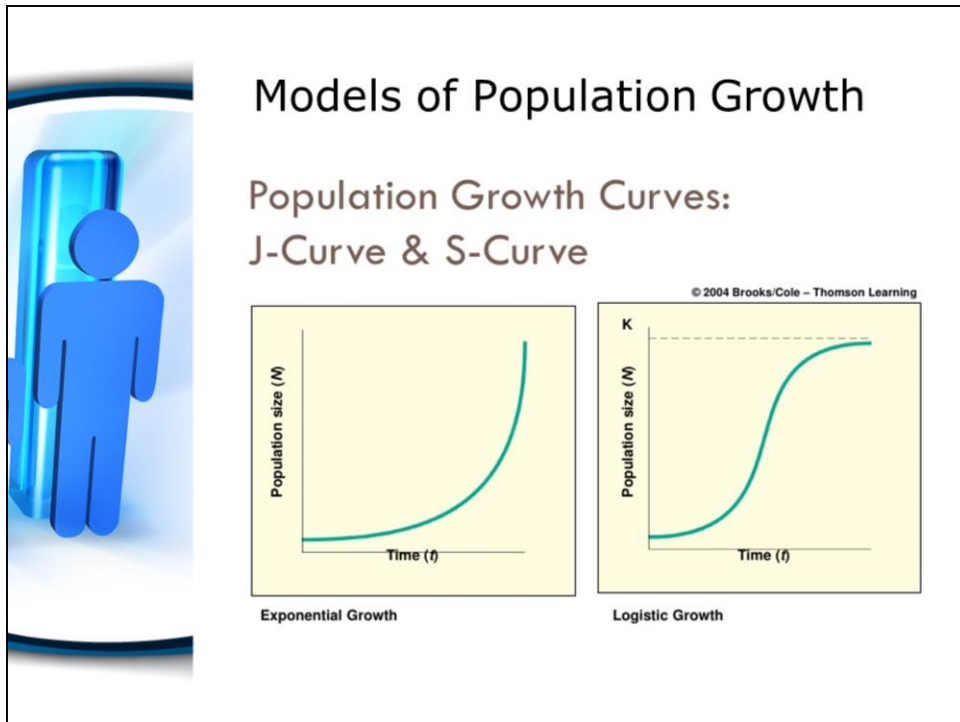
- # of people environment can support without sustaining damage





Overpopulation – there are too many people in a given area to be supported by the environment without damaging it

Underpopulation – the environment can sustain the population without damage. The population cannot or *does not* use all the available resources and maintains a high standard of living. Challenges of underpopulation arise if there are not enough people available to do the work necessary to meet the needs of the population.



The J-curve and the S-curve are two models of population growth.

The J curve models ONLY exponential growth, with population growth continuing unchecked indefinitely.

while the **S-curve** models a period of exponential growth before it tapers off and the population stabilizes again.

Malthusian Theory

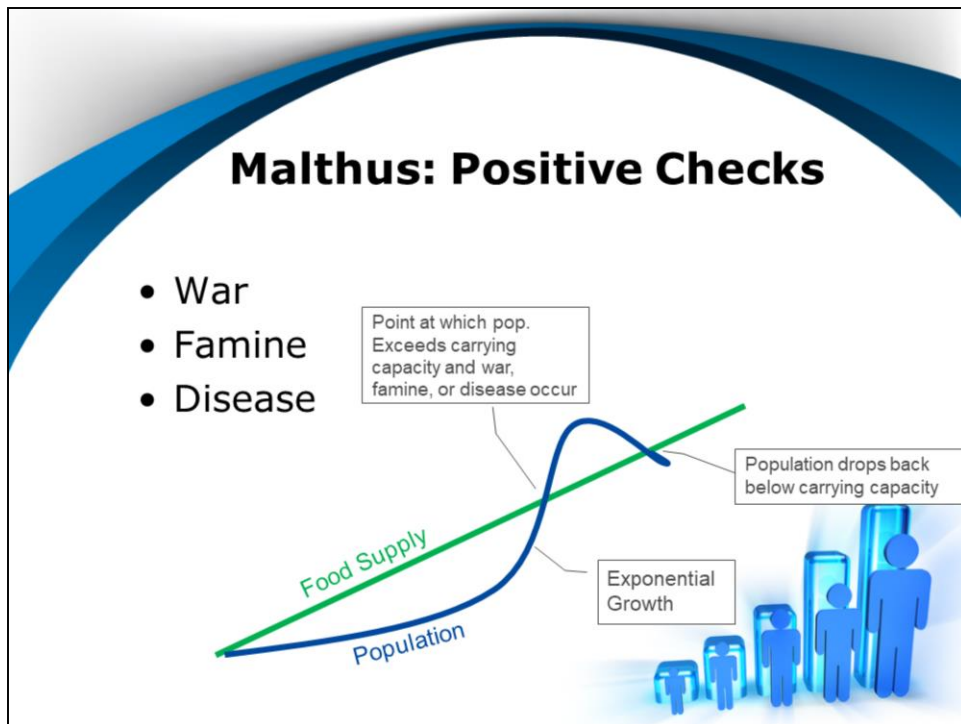
- Thomas Malthus
 - 1766-1834
 - English cleric and economist
 - Believed population growth would continue exponentially until it was “checked” (reduced) by some event.
 - Positive checks include famine, war, and disease
 - Preventative check includes “moral restraint”



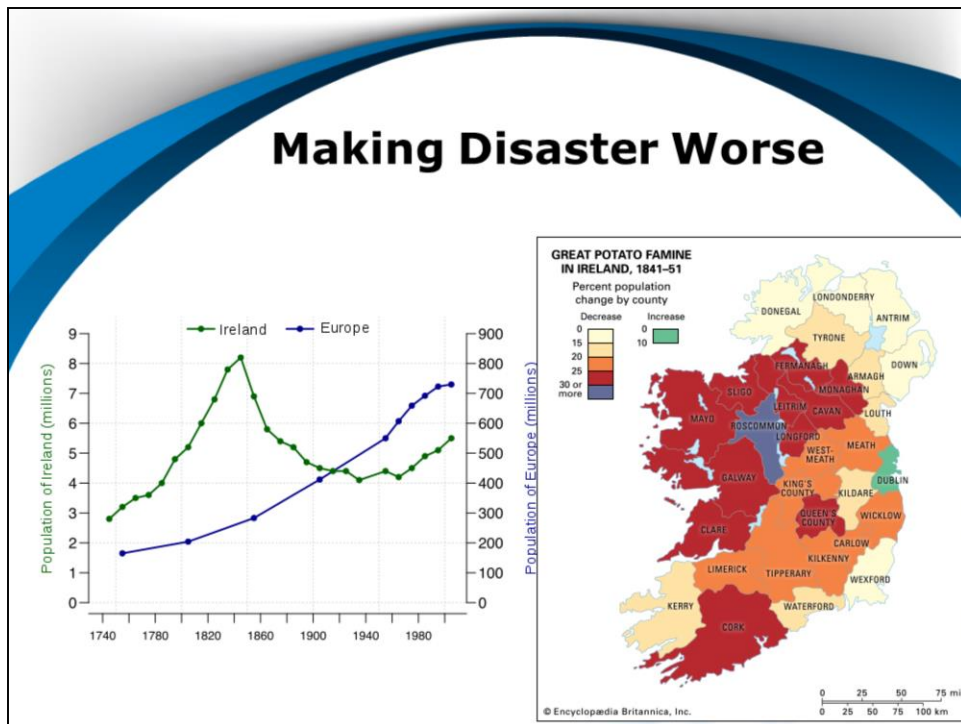
Thomas Malthus was an English cleric, demographer, and economist who lived during the early development of the industrial cities of England. He witnessed the massive influx of migrants from the rural areas as the 2nd Agricultural Revolution and England's Enclosure movement (the legal process of fencing off farm land and consolidating small farms into larger ones under a single land owner. Once enclosed, use of the land became restricted and available only to the owner, and it ceased to be common land for communal use, displacing large numbers of people from the rural areas). These two events began during the 1400s, and created a positive feedback loop – one encouraged and made the other more profitable. Combined with new food sources that the European explorers brought back from the Americas in the 1500 and 1600s, the result was a massive increase in food supply and production which resulted in a population explosion.

Thomas Malthus's idea of population growth fell on the J-curve theory – that population growth is exponential, while our ability to increase the food supply is linear. He theorized that humans had a tendency to use abundance for population growth rather than for maintaining a higher standard of living, and that once the population rose to meet the increased supply, the misery and starvation among the lower classes would continue in the “cycle of misery” – basically saying that if the poor are able to get more food, they'll just have

more kids, and therefore they will always be poor and hungry. He saw the population rising along the J-curve until it exceeded carrying capacity, which was the point at which his “Positive Checks” would intervene in the growth to reduce the population size back below carrying capacity.



Thomas Malthus considered environmental and political disasters that resulted in a decreased population size to be “positive” checks, in that they kept the population size from exceeding the carrying capacity for an extended period of time.



Beginning about 1845, the Island of Ireland (then a British colony) was hit by a fungus that destroyed their potato crops. The colonial governor and politicians responding to the crisis had read Thomas Malthus's essays and theories about population growth. These theories, based on their understanding of Malthusian Theory, said nothing should be done to help the starving Irish people because the disaster was a natural consequence of overpopulation and that the famine was a necessary "positive check" that would bring the population of Ireland back "under control" – the result was that the British barred their Irish subjects from receiving food aid and supplies from other countries. An estimated 1 million people died, and another 3 million left the island for other places, including a huge migration to the U.S. The result was that over *half* of the population of Ireland either died of starvation or emigrated to avoid starvation – and it still has not recovered today (it's about 6 million today for the entire island, compared to just over 8 million in 1840).

Notice on the map above, the only place that experienced population growth during the time period shown was Dublin, the largest city in Ireland. Also notice that the northern-most counties in Ireland experienced little to no change – this would prove to be an important divide socially, economically, and politically later as the Irish on the southern part of the island began to push back against their colonial leaders, while those in the northern-most counties remained loyal

to England (to be discussed in more detail during our culture AND political units).

Preventative Checks

- Moral restraint of the higher classes
- Eugenics



It has appeared that from the inevitable laws of our nature, some human beings must suffer from want. These are the unhappy persons who, in the great lottery of life, have drawn a blank.

(Thomas Malthus)

One way to prevent this huge population, Malthus contended, was to practice “moral restraint” as a preventative check on population growth. He felt only the higher classes would be capable of practicing this restraint, though. There was a very strong stigma in the class divide about family size – only the poor/low class had more children than they could afford to feed (changing perceptions about family size and social status that often accompanies economic development).

For those unwilling or unable to practice moral restraint, Malthus advocated **Eugenics**. Eugenics is the idea of improving human populations through selective breeding or sterilization.

Malthus believed society should limit reproduction, especially among the **poor**. Remember, Malthus lived in a time where there were many poor people moving into rapidly developing industrial cities after being forced off their land in the rural areas. They had no healthcare, little to no money, and lived in slums. He believed that the population problem was their own fault, that inequality results from laws of nature. The poor lacked “moral restraint”, and Malthus felt that to provide them with food aid would only increased their

misery because there it will keep them from reaching a positive check and prolong their suffering.

It's important, however, to see the context of Malthusian theory. During his time, innovations in English agriculture occurred eliminating jobs faster than new jobs could be created. This led to the belief among the wealthy that there was a "surplus" of workers. He had no idea of the innovations and the advancements of the Industrial Revolution that would completely change cities and way of life for the world.

Neo-Malthusians

- “neo-malthusian” is a term used for a 2nd wave of Malthusian-based population theory
 - **Neo = new**
 - **Focus on controlling population growth through contraception, particularly for poor populations**
 - **Preserve resources for themselves and their future needs**



A new wave of Malthusian-based theory that started in the late 1800s as Europe was reaching industrial maturity – think the days of thick smog and tons of pollution. This diverted the debate on population from issues of poverty and unequal access to resources, to birth control per se. The assumption of neo-malthusians is that access to commons (open space that could be used by anyone) or availability of resources would give the poor little reason to abstain from having more children.

Neo-Malthusians doubled-down on property rights to ensure that “the poor” would not infringe on the ability of the wealthy to access or use as much of the resources as they wanted/needed to run their businesses or maintain their social status. It took the responsibility of preserving resources for future use off themselves and put it on the poor people – instead of limiting their own consumption of the resources to make sure there were enough to go around and available for use in the future, they wanted the poorer people to stop having children so that there would not be more people trying to use the resources.

Today, the term neo-Malthusian is often used as a catch-all term for those who want to limit population growth as a means to ensure the preservation of

resources for future populations. It still focuses on population size rather than reducing the amount of resources being consumed.

Critics of Malthus

- Could not have foreseen the agricultural, industrial and technological advances that made it possible for population growth to slow and stabilize.
- Was looking at a small, closed system—it only applied to England and the world is not so simple.
- Did not take migration into account.



Most critics of Malthus say he couldn't see outside of his "bubble" of the dirty, overcrowded, early-industrial cities of England, or, as the saying goes, he "couldn't see the forest for the trees". He could not see the technological inventions and improvements that were happening in England and other parts of the world during his lifetime that made such massive improvements to agriculture that allowed for a massive increase in food production. He saw the displaced farm workers, but did not see or understand the impact of what had displaced them. Additionally, there is no way he could've predicted the series of inventions and modern technology that were to come which would lead to a change in way of life that would result in changing demographics and higher standard of living worldwide.



Cornucopians

- Cornucopians believe that people will find a way to solve the problem.
- They do not believe in a population doomsday.



Cornucopians think that Neo-Malthusians are “doom and gloom pessimists”, while Neo-Malthusians view Cornucopians as “unrealistic technological optimists”.

Cornucopians believe that humans will be able to use innovation and invention to create techniques or technology that will be able to solve the resource problems - that there are always alternatives, and will be new technology to come to meet the needs of the people. The term “cornucopian” is derived from the image associated with good harvest - the cornucopia, or “the horn of plenty”. In classical mythology, it was a horn of the goat Amalthea containing food, drink, and resources in endless supply.



Ester Boserup

- A “cornucopian.”
- Population growth drives technological advances.
- In turn, these advances raise food production and improve economic development.



Ester Boserup (1910–1999) was a 20th century economist who specialized in the economics and development of agriculture in her work with the United Nations.

In 1965, she published an essay, *‘The Conditions of Agricultural Growth: The economics of agrarian change under population pressure’*, Boserup challenged Malthus’s conclusion that the size of the human population is limited by the amount of food it can produce. She suggested that food production can, and will, increase to match the needs of the population. Using her experiences working in India, which at the time was an LDC with a rapidly growing population, Boserup argued that the threat of starvation and the challenge of feeding more children motivated people to improve their farming techniques and to invent new technologies in order to increase crop yields and food production. She called this change “agricultural intensification” and provided the example of a farmer with four fields. Based on his current family size, the farmer was only using three of the four fields to grow food because the 4th field was too dry. If he had two more children, however, he would need to grow more food (additional pressure to produce more), and it would prompt him to invest the resources necessary to dig irrigation canals to bring water into the fields so that they could be farmed. He might also be willing to spend more money on new or different types of seeds that would grow in drier soil.

He would change the way he farms in order to meet the needs of his larger family.



Role of Technology

- Machinery
 - Change landscape to create more farmland
 - Terraces
 - Increase distance produce can be shipped
- Fertilizer and pesticides
 - Science-created, organic vs. chemical
- Digital Technology

The technology and innovations of the Industrial Revolution vastly increased the tools humans had to use to create more food. Cornucopians' views of population theory are that humans will be able to innovate new ways to increase the food supply.

Examples of ways in which technology has increased the food supply:

Heavy equipment can be used to literally move mountains – or at least carve terraces into the side of them – that expand the amount of farmland on the mountainsides. Technology such as the steam engine allowed food to be shipped large distances, and when refrigeration became available, it allowed for global trade of perishable foods (food products that go bad quickly unless they are chilled).

Fertilizer (add nutrients to the soil) and pesticides (kill harmful bugs) have been around for millennia in their organic form (i.e., animal and plant waste to add organic material to the soil), but advancements of the industrial revolution lead to a more science-based approach to agriculture that resulted in *chemical* (non-organic) fertilizers and pesticides that were even more impactful

Today, digital technology such as GPS and GIS are being used to analyze field health and to apply “cures” to the soil exactly where they are needed – such as more targeted irrigation or adding soil to plots experiencing erosion of the top soil.